

Cloud-Robust Field-Scale Crop Intelligence with Sentinel-1/Sentinel-2 Fusion: An AgriSense Study Linking Karnal Operations, Haridwar Phenology and AgriFieldNet Crop Labels

Atharva N.¹, [Co-author 2]¹ and [Faculty Guide]¹

¹Department of Computer Engineering, K. J. Somaiya Institute of Technology, Mumbai, India
Email: [corresponding author email]

Abstract—Crop-type mapping and phenology inference in monsoon-influenced smallholder agriculture are constrained by cloud gaps, heterogeneous fields and uneven ground supervision. This paper presents a provenance-aware AgriSense study that links three complementary components without claiming ownership of externally created datasets. The Karnal operational layer is derived by the present authors from Copernicus Sentinel-1 and Sentinel-2 observations processed in Google Earth Engine. The Haridwar growth-stage dates are credited to Akash Kumar and Siddhartha Khare, who collected and published two seasons of wheat field observations at KVK Dhanauri; AgriSense uses those dates as externally sourced stage anchors, groups the ten published stages into five broader classes, and performs a separate secondary analysis with independently extracted S1/S2 features. AgriFieldNet is credited to Radiant Earth Foundation and IDinsight: its ground-survey labels cover 7,081 fields in total, including 5,551 training fields across 13 classes; AgriSense reconstructs the 5,551 labelled training fields and extends them with a leakage-safe multi-temporal S1/S2 extraction, but does not report a crop-classification score until the final spatial cross-validation is complete. In the Haridwar secondary analysis, the selected S1+S2 Random Forest obtains macro-F1 0.493 under leave-one-season-out evaluation, compared with 0.461 for the selected S2-only model and 0.429 for S1-only. Under 60% additional optical masking, fusion retains macro-F1 0.424 while the S2-only model falls to 0.267. The contribution is therefore not a claim of first-use of S1/S2 crop mapping, Haridwar phenology, or AgriFieldNet classification; it is the explicit integration of credited data sources, sensor-missingness auditing, cross-dataset supervision rules, cloud-gap robustness analysis and deployment in one field-scale application architecture.

Index Terms—Sentinel-1, Sentinel-2, crop classification, phenology, SAR-optical fusion, AgriFieldNet, cloud robustness, data provenance, field-scale remote sensing, AgriSense.

I. INTRODUCTION

Field-scale crop intelligence is a temporal problem rather than a single-image problem. Crop type is expressed through seasonal trajectories, growth stages change over time, and optical observation availability varies with cloud, haze, revisit geometry and field size. Sentinel-2 provides spectrally rich optical observations [2], whereas Sentinel-1 contributes weather-independent C-band SAR measurements [1]. Google Earth Engine provides the cloud-scale processing environment used in the AgriSense workflow [3].

The literature already establishes that radar-optical fusion can improve crop mapping. Rao, Zhou, Bhattarai, Srivastava, Singh, Poonia, Lobell and Jain mapped four smallholder crop types in eastern India using Sentinel-1, Sentinel-2 and PlanetScope and reported 85% accuracy for the three-sensor combination [4]. Tufail, Ahmad, Javed and Ahmad combined Sentinel-1 and Sentinel-2 time series with Random Forest and reported approximately 97% overall accuracy in their Pakistan study [5]. Woźniak and co-authors showed that Sentinel-1 time-series phenological descriptors can strengthen crop classification [6]. These studies mean that AgriSense should not be positioned as the first SAR-optical crop-classification framework.

Field-level phenology estimation also predates this work. Sawant, Mohite and Pappula combined field observations with Sentinel-2 time series for wheat growth monitoring in India [7]. More recently, Kumar and Khare conducted a two-season wheat phenology experiment at KVK Dhanauri, Haridwar using in-situ observations, an infrared-enabled PhenoCam, Sentinel-2 and PlanetScope [8]. Their study reported phenophase errors in days and released associated analysis resources and observations on Figshare [9]. AgriSense therefore treats the Haridwar observations as credited external evidence, not as original field data collected by the present authors.

The third component, AgriFieldNet, is likewise an established public benchmark. Radiant Earth Foundation and IDinsight published the AgriFieldNet Competition Dataset with ground reference data collected by IDinsight's Data on Demand team and dataset curation/publication by Radiant Earth Foundation [10]. The Zindi challenge already produced strong solutions; the first-place model by Muhamed Tuo, Caleb Emelike and Taiwo Ogundare combined one U-Net with eight gradient-boosting models [11]. Accordingly, this paper does not claim to be the first to classify AgriFieldNet. Its distinct contribution is to reconstruct labelled fields from the raster chips, attach a multi-temporal S1/S2 sequence, preserve sensor missingness and evaluate the resulting architecture alongside a credited Haridwar phenology reanalysis and the Karnal operational layer.

II. STUDY POSITIONING AND CONTRIBUTIONS

The paper asks a narrower and more defensible question: can one field-time fusion architecture support crop-type and phenology research across datasets with different supervision, while clearly separating original observations, externally sourced labels and AgriSense-derived products? Four contributions are evaluated. First, a common field-time representation stores optical indices, Sentinel-1 backscatter and quality flags without hiding missingness. Second, the Haridwar stage chronology published by Kumar and Khare [8] is re-used with explicit attribution for a new S1/S2 five-class secondary analysis and optical-gap stress test. Third, AgriFieldNet’s original crop labels [10] are transformed into field polygons and extended with a planned 21-period S1/S2 temporal stack; no phenology labels are invented. Fourth, the same outputs are connected to the role-specific AgriSense application [12].

This framing deliberately separates novelty from source ownership. Kumar and Khare remain the source authors for the Haridwar field observations and stage dates; Radiant Earth Foundation and IDinsight remain the source creators/curators of AgriFieldNet; ESA/Copernicus remain the source of Sentinel observations. The AgriSense contribution is the secondary processing, fusion, validation design, robustness experiments and software integration performed on top of those sources.

III. DATA SOURCES, PROVENANCE AND ATTRIBUTION

Component	Original source / credit	Supervision actually available	Role of AgriSense in this paper
Karnal	Copernicus Sentinel-1/2 [1], [2]; processed by present authors in Earth Engine [3]	Satellite/quality time series; no independent crop-stage ground truth used for final claims	Operational field-time layer, natural optical-gap audit, application deployment
Haridwar	Wheat stage dates and site evidence from Akash Kumar & Siddhartha Khare [8], with associated open resources [9]	Ten published stage dates per season; two seasons; AgriSense daily/group labels are derived from those anchors	Secondary S1/S2 classification, transition analysis and cloud-gap robustness; not original field observation collection
AgriFieldNet	Radiant Earth Foundation & IDinsight [10]; prior challenge solutions include Tuo, Emelike & Ogundare [11]	Observed crop type for 7,081 fields total; 5,551 training fields; 13 classes; no field-date phenology labels	Reconstruct labelled fields, attach multi-temporal S1/S2 features and run spatial validation; final score pending

TABLE I. Data provenance, ownership and the specific use made by AgriSense.

A. Karnal Operational Layer

The Karnal layer is an AgriSense-derived operational dataset rather than a published ground-truth benchmark. It contains 4,311 field/sample locations over ten composite starts from June to October 2025. Sentinel-2 optical validity varies strongly across the season, whereas Sentinel-1 is available at every location-period in the current table. These data support an engineering claim about observation continuity and deployment, not a validated crop-classification accuracy claim. Current pseudo/derived crop or stage targets in the application are excluded from the paper’s final agronomic accuracy claims.

B. Haridwar Phenology Evidence

Kumar and Khare [8] collected the Haridwar wheat observations at KVK Dhanauri and published ten stage dates for each of the 2023–24 and 2024–25 seasons, including Germination, Leaf Development, Tillering, Stem Elongation, Booting, Heading, Flowering, Development of Fruit, Ripening and Senescence/Harvest. The present study does not claim those field observations. Instead, it transcribes the published stage chronology, maps the ten stages into five broader groups—Establishment, Vegetative, Reproductive, Grain Filling and Senescence/Harvest—and independently extracts/uses Sentinel-1 and Sentinel-2 features for a secondary machine-learning analysis. The five-class grouping and the daily midpoint interpolation between adjacent published dates are AgriSense design choices, not labels originally reported by Kumar and Khare.

This distinction is important because Kumar and Khare [8] evaluated phenological dates using PhenoCam, Sentinel-2 and PlanetScope and reported errors in days. The AgriSense results below instead evaluate a five-class sensor-fusion problem with leave-one-season-out testing. Their published MAE values and our macro-F1 values answer different questions and should not be compared as if they were the same benchmark.

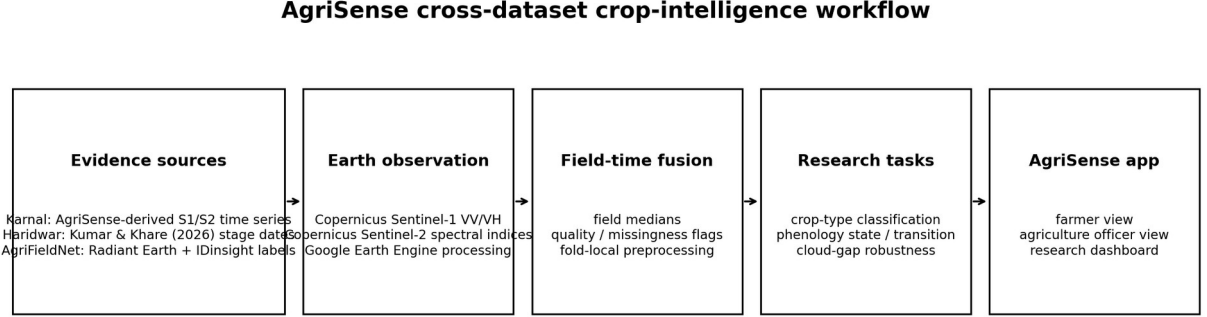
C. AgriFieldNet Crop Labels

Radiant Earth Foundation and IDinsight [10] describe AgriFieldNet as 1,217 Sentinel-2 image/label chips containing 7,081 fields, split into 5,551 training and 1,530 test fields. The crop labels originate from ground survey, and the official dataset contains 13 classes including fallow and 12 crop types. Ground reference collection is credited to IDinsight’s Data on Demand team, while Radiant Earth Foundation curated and published the training dataset. The associated ECAAS initiative was funded by the Bill & Melinda Gates Foundation and implemented by Tetra Tech [10].

AgriSense converts the training-label rasters to one polygon per global field identifier, checks for label conflicts, and then prepares multi-temporal Sentinel-1/Sentinel-2 field summaries. This temporal reconstruction

is an additional analysis performed by the present study; it is not part of the original single-composite AgriFieldNet release. The temporal window will be frozen against source metadata before final submission. Because AgriFieldNet contains crop-type labels but no observed per-field phenology stages, it is used only for crop classification.

IV. METHODOLOGY



Original data ownership is retained; AgriSense contributions are transformations, secondary analyses and deployment.

Figure 1. Cross-dataset AgriSense workflow with explicit data-source attribution.

A. Common Field-Time Representation

Sentinel-2 surface reflectance is summarized with vegetation/moisture indices such as NDVI, NDMI, NDWI, EVI and SAVI. Sentinel-1 is represented by VV and VH backscatter, difference/ratio features and acquisition-count diagnostics. Optical-valid and SAR-valid indicators are preserved as features. Imputation and scaling are fitted inside each training fold to avoid leakage. The framework therefore distinguishes a missing optical observation from a vegetation observation with a genuinely low index value.

B. Haridwar Secondary Classification and Validation

For each season, satellite acquisitions within the published phenology window are assigned to the nearest interval defined by midpoints between adjacent Kumar–Khare stage dates [8]. This produces 59 labelled acquisition rows across both seasons after grouping to five stage classes. Three feature families are evaluated: S2-only, S1-only and S1+S2 fusion. Models are compared using leave-one-season-out (LOSO) validation, so one full season is withheld during each test. Macro-F1 is the primary metric because the derived five-class table is highly imbalanced, including only one Senescence/Harvest row. Accuracy, balanced accuracy, Cohen's kappa and MCC are retained as secondary metrics.

Cloud-gap robustness is tested by adding controlled optical masking after model selection. The fusion model retains SAR features when S2 features are masked, whereas the S2-only model receives only the remaining optical information. This experiment measures graceful degradation under missing optical observations rather than claiming that SAR is universally more accurate than optical data.

C. AgriFieldNet Temporal Extension and Spatial Validation

The AgriFieldNet converter polygonizes `field\_ids.tif`, joins the corresponding crop labels, detects cross-chip conflicts, dissolves fragments by global field_id and preserves raster-faithful boundaries. The QA stage recovers the expected 5,551 labelled training fields and 13 crop classes without silently resolving conflicting labels. Periodic extraction keeps Sentinel-1 ascending and descending passes separate and summarizes field observations by medians. A 21-bin, ten-day design yields an expected 116,571 field-period rows before any structural exclusions.

The planned model benchmark compares S2 static summaries, S2 temporal features, S1 temporal features, fused S1/S2 temporal features and compact temporal descriptors. Spatial grouping is used where geometry support permits, and the primary score is macro-F1, followed by balanced accuracy, weighted-F1, Cohen's kappa and MCC. The original AgriFieldNet challenge already demonstrated that the dataset is classifiable [11]; therefore, the final paper will report our temporal-fusion results only after the spatial run completes rather than substituting challenge scores or pseudo-label results.

V. RESULTS

A. Haridwar LOSO Secondary Analysis

The following results are AgriSense reanalysis results, not values reported by Kumar and Khare [8]. The selected S1+S2 fusion model is Random Forest with accuracy 0.644, balanced accuracy 0.504 and macro-F1 0.493. The selected S2-only RBF SVM obtains accuracy 0.559 and macro-F1 0.461. The selected S1-only Random Forest also obtains accuracy 0.644 but macro-F1 0.429. The difference between identical overall accuracy and lower macro-F1 for S1-only shows why class-balanced metrics are necessary in this small, imbalanced secondary dataset.

Feature family	Selected model	Accuracy	Balanced acc.	Macro-F1	Kappa
S2 only	RBF SVM	0.559	0.468	0.461	0.416
S1 only	Random Forest	0.644	0.452	0.429	0.484
S1 + S2	Random Forest	0.644	0.504	0.493	0.506

TABLE II. AgriSense LOSO reanalysis of Haridwar stage groups; source stage dates are credited to Kumar and Khare [8].

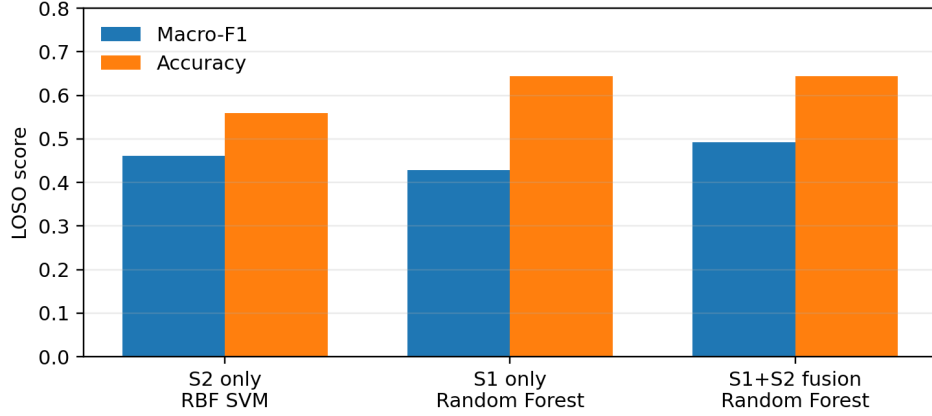


Figure 2. AgriSense LOSO performance for the selected model in each Haridwar sensor family.

B. Cloud-Gap Robustness

The Haridwar acquisition calendar contains natural optical gaps: usable S2 fractions are approximately 60.5% in 2023–24 and 73.3% in 2024–25 in the AgriSense extraction. In the additional masking experiment, the fusion Random Forest declines from macro-F1 0.493 to 0.424 at 60% extra optical masking, while the selected S2-only RBF SVM declines from 0.461 to 0.267. The absolute values depend on the small two-season dataset, but the relative degradation supports the engineering rationale for retaining Sentinel-1 when optical observations disappear.

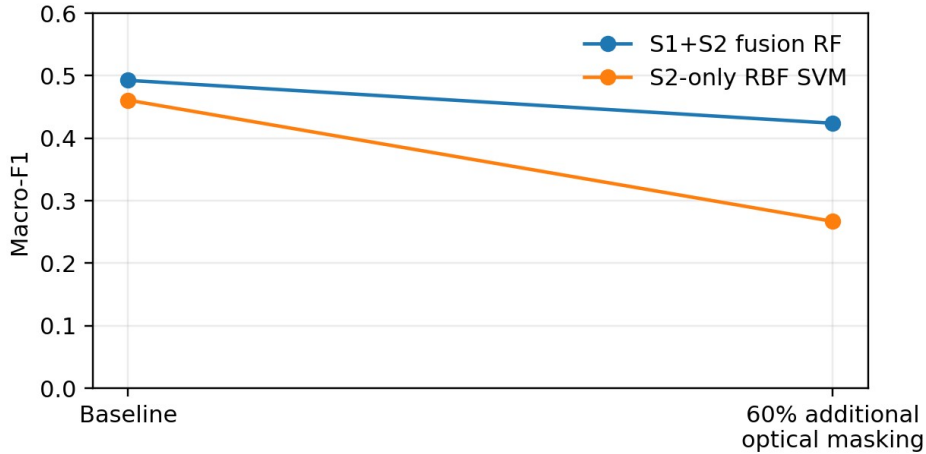


Figure 3. Haridwar robustness in the AgriSense reanalysis: baseline versus 60% additional optical masking.

C. Karnal Natural Missingness

Karnal provides a separate operational observation rather than a supervised accuracy benchmark. Across 4,311 locations, optical validity is 100% on 1 June 2025, 31.2% on 17 June, 1.6% on 3 July, 0% on 19 July and 4 August, 25.7% on 20 August, approximately 100% in early and late September/early October, and 41.9% on 23 October. Sentinel-1 validity is 100% in every period. These are derived AgriSense coverage statistics based on Copernicus observations [1], [2], not field-collected agronomic labels.

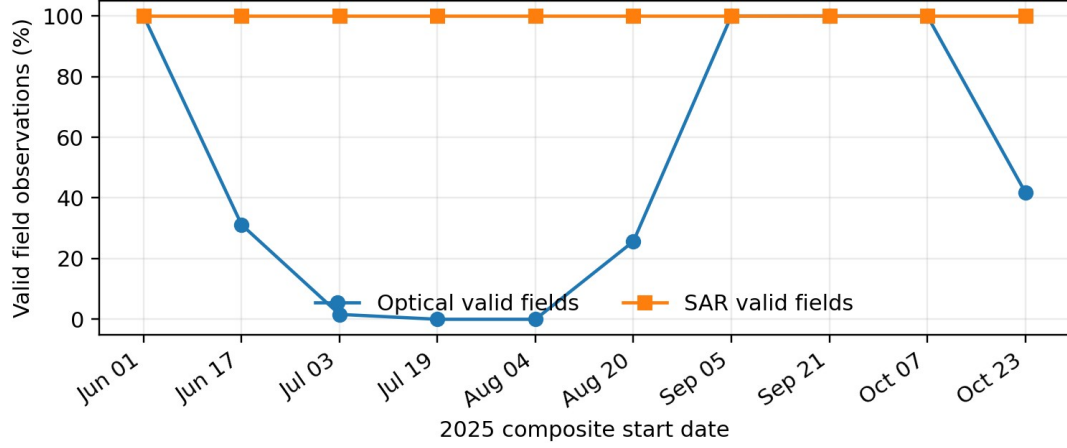


Figure 4. Karnal observation availability in the AgriSense derived field-time table.

D. AgriFieldNet Audit Status

The completed AgriFieldNet data-engineering audit recovers 5,551 unique labelled training fields, 13 crop classes and no unresolved crop-label conflicts in the converted field set. The planned 21-period design corresponds to 116,571 field-period rows when every field is represented in every period. These are structural pipeline results; they are not crop-classification performance results. The paper intentionally leaves the final AgriFieldNet model score unreported until spatially grouped cross-validation and rare-class fold support are complete.

VI. DISCUSSION AND COMPARISON WITH PRIOR WORK

The Haridwar experiment should be interpreted as a secondary extension of Kumar and Khare's phenology study, not a replacement for it. Kumar and Khare [8] used in-situ observations as the primary validation source and compared PhenoCam, Sentinel-2 and PlanetScope phenological dates. AgriSense adds Sentinel-1 and reframes the problem as five-class S1/S2 stage-state classification plus optical-gap robustness. Because the target definitions and metrics differ, their day-based MAE values cannot be directly ranked against our macro-F1 values.

Likewise, the crop-classification literature already demonstrates strong multisensor performance. Rao and co-authors [4] showed that higher-resolution multi-sensor imagery can help smallholder crop mapping in India, while Tufail and co-authors [5] showed strong performance from S1/S2 fusion in Pakistan. Woźniak and co-authors [6] demonstrated the value of SAR phenology descriptors at much larger scale. AgriSense contributes a different evaluation emphasis: explicit sensor missingness, cross-dataset provenance, season-held-out testing and a deployment layer that exposes the resulting state to stakeholder-specific interfaces.

AgriFieldNet also has prior benchmark solutions, including the first-place model by Tuo, Emelike and Ogundare [11]. The research contribution here is therefore not 'classification of AgriFieldNet' by itself. Instead, AgriSense reconstructs fields from the original raster labels, adds a temporal S1/S2 observation sequence and evaluates whether temporal radar-optical information improves spatially held-out crop classification relative to optical-only representations. That comparison remains pending and must be completed before any claim of improved AgriFieldNet accuracy is made.

VII. NOVELTY BOUNDARY, LIMITATIONS AND VALIDITY

The novelty boundary is intentionally narrow. This study does not claim: (i) the first use of Sentinel-1 and Sentinel-2 for crop classification; (ii) the first field-level wheat phenology study in India; (iii) the first analysis of the Haridwar KVK experiment; or (iv) the first AgriFieldNet crop classifier. The distinctive contribution is the provenance-aware cross-dataset integration of a Karnal operational layer, a credited Haridwar phenology chronology extended with an S1/S2 secondary analysis, and a temporal S1/S2 reconstruction of AgriFieldNet, all evaluated under explicit missing-data and leakage controls and exposed through one application architecture.

Several limitations constrain generalization. The Haridwar secondary dataset covers only two seasons at one experimental site, and the five grouped classes are derived from published anchor dates rather than direct labels for every satellite acquisition. Senescence/Harvest has extremely low support. The Karnal layer lacks independent crop/stage ground truth suitable for publication-grade accuracy assessment, so it is used only for coverage and deployment evidence. AgriFieldNet contains crop labels but no observed field-date phenology;

final crop-classification metrics remain pending. Sentinel-2 bands with native 20 m resolution do not gain new spatial information when reduced on a 10 m grid. Finally, the AgriFieldNet temporal window must be frozen against source metadata before the definitive experiment.

VIII. CONCLUSION AND FUTURE WORK

The revised AgriSense study shows how remote-sensing research can be integrated without obscuring who created the underlying observations. The strongest completed quantitative evidence comes from a clearly labelled secondary analysis of Kumar and Khare's Haridwar stage chronology: S1/S2 fusion provides the highest macro-F1 among the selected sensor families and degrades less sharply than S2-only inference when optical observations are deliberately removed. The Karnal layer independently demonstrates why cloud-resilient continuity matters in deployment. AgriFieldNet supplies the multi-crop ground truth needed for a larger crop-type test, but no final score is claimed until the spatial experiment is complete.

Before submission, future work will complete AgriFieldNet spatial cross-validation and sensor ablation, archive the exact temporal-window manifest and code release, add uncertainty intervals to Haridwar metrics, and test cross-region transfer only where crop calendars and label semantics are compatible. Direct field phenology observations in Karnal would be required before presenting Karnal phenology accuracy as independently validated.

ACKNOWLEDGEMENT AND ATTRIBUTION STATEMENT

The present authors acknowledge that the Haridwar field observations and published wheat stage chronology were created by Akash Kumar and Siddhartha Khare at IIT Roorkee [8], with open analysis resources made available through Figshare [9]. Their paper acknowledges Krishi Vigyan Kendra (KVK) Dhanauri, Haridwar, for the experimental site and specifically thanks Shri Umesh Saxena and Dr. Purushottam Kumar for support during field observations; we preserve that credit here. The present authors did not collect those original Haridwar field observations.

The AgriFieldNet Competition Dataset is credited to Radiant Earth Foundation and IDinsight [10]. Ground reference data were collected by IDinsight's Data on Demand team; Radiant Earth Foundation curated and published the training dataset. The ECAAS initiative supporting the dataset was funded by the Bill & Melinda Gates Foundation and implemented by Tetra Tech [10]. We also acknowledge the prior AgriFieldNet competition work of Muhamed Tuo, Caleb Emelike and Taiwo Ogundare [11].

Copernicus Sentinel-1 and Sentinel-2 data are provided through the European Commission/European Space Agency programme [1], [2]. Google Earth Engine is acknowledged as the geospatial processing platform [3]. The AgriSense software and derived processing code used in this secondary analysis are maintained in the project repository [12].

DATA AND CODE AVAILABILITY

Haridwar source observations and associated analysis resources are available from the data link reported by Kumar and Khare [8] and the Figshare record [9]. AgriFieldNet is available from the Radiant Earth/Source Cooperative release under DOI 10.34911/rdnt.wu92p1 [10]. AgriSense source code, derived Karnal tables and the Haridwar/AgriFieldNet processing notebooks are maintained at the project repository [12]. A frozen release containing the exact paper figures, manifests and final AgriFieldNet outputs should be archived before final publication.

REFERENCES

- [1] R. Torres, P. Snoeij, D. Geudtner, D. Bibby, M. Davidson, E. Attema, P. Potin, B. Rommen, N. Floury, M. Brown, I. Navas Traver, P. Deghaye, B. Duesmann, B. Rosich, N. Miranda, C. Bruno, M. L'Abbate, R. Croci, A. Pietropaolo, M. Huchler and F. Rostan, "GMES Sentinel-1 mission," *Remote Sensing of Environment*, vol. 120, pp. 9-24, 2012, doi:10.1016/j.rse.2011.05.028.
- [2] M. Drusch, U. Del Bello, S. Carlier, O. Colin, V. Fernandez, F. Gascon, B. Hoersch, C. Isola, P. Laberinti, P. Martimort, A. Meygret, F. Spoto, O. Sy, F. Marchese and P. Bargellini, "Sentinel-2: ESA's Optical High-Resolution Mission for GMES Operational Services," *Remote Sensing of Environment*, vol. 120, pp. 25-36, 2012, doi:10.1016/j.rse.2011.11.026.
- [3] N. Gorelick, M. Hancher, M. Dixon, S. Ilyushchenko, D. Thau and R. Moore, "Google Earth Engine: Planetary-scale geospatial analysis for everyone," *Remote Sensing of Environment*, vol. 202, pp. 18-27, 2017, doi:10.1016/j.rse.2017.06.031.
- [4] P. Rao, W. Zhou, N. Bhattarai, A. K. Srivastava, B. Singh, S. Poonia, D. B. Lobell and M. Jain, "Using Sentinel-1, Sentinel-2, and Planet Imagery to Map Crop Type of Smallholder Farms," *Remote Sensing*, vol. 13, no. 10, 1870, 2021, doi:10.3390/rs13101870.
- [5] R. Tufail, A. Ahmad, M. A. Javed and S. R. Ahmad, "A machine learning approach for accurate crop type mapping using combined SAR and optical time series data," *Advances in Space Research*, vol. 69, pp. 331-346, 2022, doi:10.1016/j.asr.2021.09.019.
- [6] E. Woźniak, M. Rybicki, W. Kofman, S. Aleksandrowicz, C. Wojtkowski, S. Lewiński, J. Bojanowski, J. Musiał, T. Milewski, P. Slesiński and A. Łączyński, "Multi-temporal phenological indices derived from time series Sentinel-1 images to country-wide crop

- classification," *International Journal of Applied Earth Observation and Geoinformation*, vol. 107, 102683, 2022, doi:10.1016/j.jag.2022.102683.
- [7] S. A. Sawant, J. D. Mohite and S. Pappula, "Integration of human participatory sensing and archives of remote sensing observations for field level crop phenology estimation," *International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences*, vol. XLII-4, pp. 547-550, 2018, doi:10.5194/isprs-archives-XLII-4-547-2018.
- [8] A. Kumar and S. Khare, "Evaluation of PlanetScope and Sentinel-2 derived wheat crop phenology using infrared-enabled PhenoCam and field observations," *Ecological Informatics*, vol. 96, 103821, 2026, doi:10.1016/j.ecoinf.2026.103821.
- [9] A. Kumar and S. Khare, Figshare data and analysis resources accompanying "Evaluation of PlanetScope and Sentinel-2 derived wheat crop phenology using infrared-enabled PhenoCam and field observations," version 2, 2026, doi:10.6084/m9.figshare.30162856.v2.
- [10] Radiant Earth Foundation and IDinsight, "AgriFieldNet Competition Dataset," Version 1.0, Radiant MLHub / Source Cooperative, 2022, doi:10.34911/rdnt.wu92p1.
- [11] M. Tuo, C. Emelike and T. Ogundare, "AgriFieldNet Model for Crop Detection from Satellite Imagery," Version 1.0, Radiant MLHub, first-place AgriFieldNet India Challenge solution, doi:10.34911/rdnt.k2ft4a.
- [12] AgriSense Project Team, "AgriSense: Satellite AI Crop and Irrigation Advisory Platform," GitHub repository, 2026. Available: <https://github.com/Atharva-cyber849/AgriSense> (accessed 23 September 2026).